

Adrián Pabón Mendoza

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Target Role

AI Engineer (LLM/RAG & Computer Vision) | Electronics Engineer (Cum Laude) | Python, PyTorch, FastAPI, SQL, Docker, AWS/Azure/OCI

Professional Summary

Electronics Engineer and AI Engineer focused on building production AI systems (Computer Vision and LLM/RAG). Scaled NAIA from a 500-user pilot to 13,000+ users by integrating OpenAI LLMs, Redis state management, and Microsoft Graph/Azure services. Delivered computer vision detection improvements reducing manual camera review time by **80%**. End-to-end experience: data and evaluation, APIs, deployment, and monitoring.

Skills

ML/LLM: PyTorch, scikit-learn, Hugging Face, fine-tuning, prompt tuning, embeddings, RAG, prompt engineering, evaluation.

Computer Vision: Training/inference pipelines, dataset curation, labeling tools (Roboflow), model deployment.

Agents: LangChain/LangGraph, tool-calling, routing.

Backend: Python, FastAPI/Flask, REST APIs, Redis, SQL.

MLOps/DevOps & Cloud: Git, Docker, AWS (EC2/S3/RDS, Lambda), Azure, Oracle Cloud, IBM Cloud, Terraform.

Experience

Double V Partners

Senior Data and AI Engineer

Remote, Colombia

Feb 2026 – Present

- Leading the migration of core banking procedures to GenAI-optimized microservices, reducing technical debt and improving system scalability through event-driven architectures.
- Designing and implementing serverless architectures on AWS (Lambda, Step Functions, Bedrock) for large-scale data processing, integrating foundation models to automate financial workflows.
- Orchestrating secure, scalable Generative AI solutions ensuring compliance with banking regulations via Amazon SageMaker and data governance services.

Optimize IT SAS

AI Product Specialist / AI Engineer

Barranquilla, Colombia

Jul 2024 – Feb 2026

- Improved theft detection in micro-market systems using Computer Vision, reducing manual camera review time by **80%**.
- Trained and evaluated ML models for production use cases (classification/regression), tracking key metrics (**F1, precision, recall, MAE**) and validating with hold-out sets.
- Delivered an LLM/RAG API with embeddings + vector search, iterating on chunking strategy, prompt templates, and retrieval parameters to improve grounded responses in internal workflows.
- Automated database backup workflows on OCI (scheduling, validation, notifications) with serverless functions, improving reliability, traceability, and reducing manual intervention.

Universidad del Norte

AI Researcher & Developer

Barranquilla, Colombia

Jan 2024 – Nov 2025

- Reduced administrative response time from hours/days to **3–5 seconds** by deploying NAIA (13,000+ users), enabling self-service for routine queries.
- Designed a modular monolith integrating OpenAI LLMs, Redis state, and external services via Azure; implemented secure access through Microsoft Graph API (email/calendar/authentication).
- Implemented RAG pipelines (chunking, embeddings, retrieval, reranking when applicable) and defined evaluation sets to reduce hallucinations and improve answer relevance.

Featured Projects

AI Agent Implementation for Medical Call Center

2026

Intelligent voice agents to improve customer care experience at a clinical laboratory.

- Executed outbound campaigns for a medical call center.
- Implemented **Tool-Calling** logic for real-time appointment and lab result validation, connecting the agent directly to the medical center's databases.

- Impact: **50%** reduction in call center agents' time spent on routine calls, freeing them for higher-impact interactions.

NAIA — Nimble Artificial Intelligence Assistant

2024–2025

Portfolio link | Multi-role assistant with LLM/RAG and integrations.

- Architecture: OpenAI LLMs + Redis + Azure services + Microsoft Graph API integration.
- Scale: 13,000+ users; definition of operational flows and monitoring for production use.
- Impact: conversational responses in **3–5 seconds** (web architecture baseline).

Theft Detection (Computer Vision)

2024

Production CV pipeline for micro-market security.

- Data: dataset curation and labeling; model training/validation with CV metrics (**mAP/F1**) and hold-out evaluation.
- Impact: **80%** reduction in manual camera review time.

Publications & Research

- **NAIA: A Robust AI Framework for Academic Assistance** — *Systems* (2025).
- **Architectures for AI-Powered Virtual Assistants** — *IEEE TEMSCON LATAM* (2025).
- **NAIA: A Multi-Technology Virtual Assistant** — *IEEE Access* (2025).
- **Itinerary Assignment Optimization** — *IEEE COLCOM* (2024).

Education

Universidad del Norte

Barranquilla, Colombia

B.Eng. in Electronics Engineering (Cum Laude)

2020 – 2025

Certifications & Recognition

- **Databricks Fundamentals** (2025) | **NVIDIA: Applications of AI for Anomaly Detection** (2025) | **NVIDIA: Generative AI with Diffusion Models** (2025)
- **Speaker:** Cátedra Europa 2024 (“Computer Vision: A World of Possibilities”). | **Recognition:** Best Science Project (Colciencias, 2019).

Languages

Spanish (Native) | English (Professional / C1)